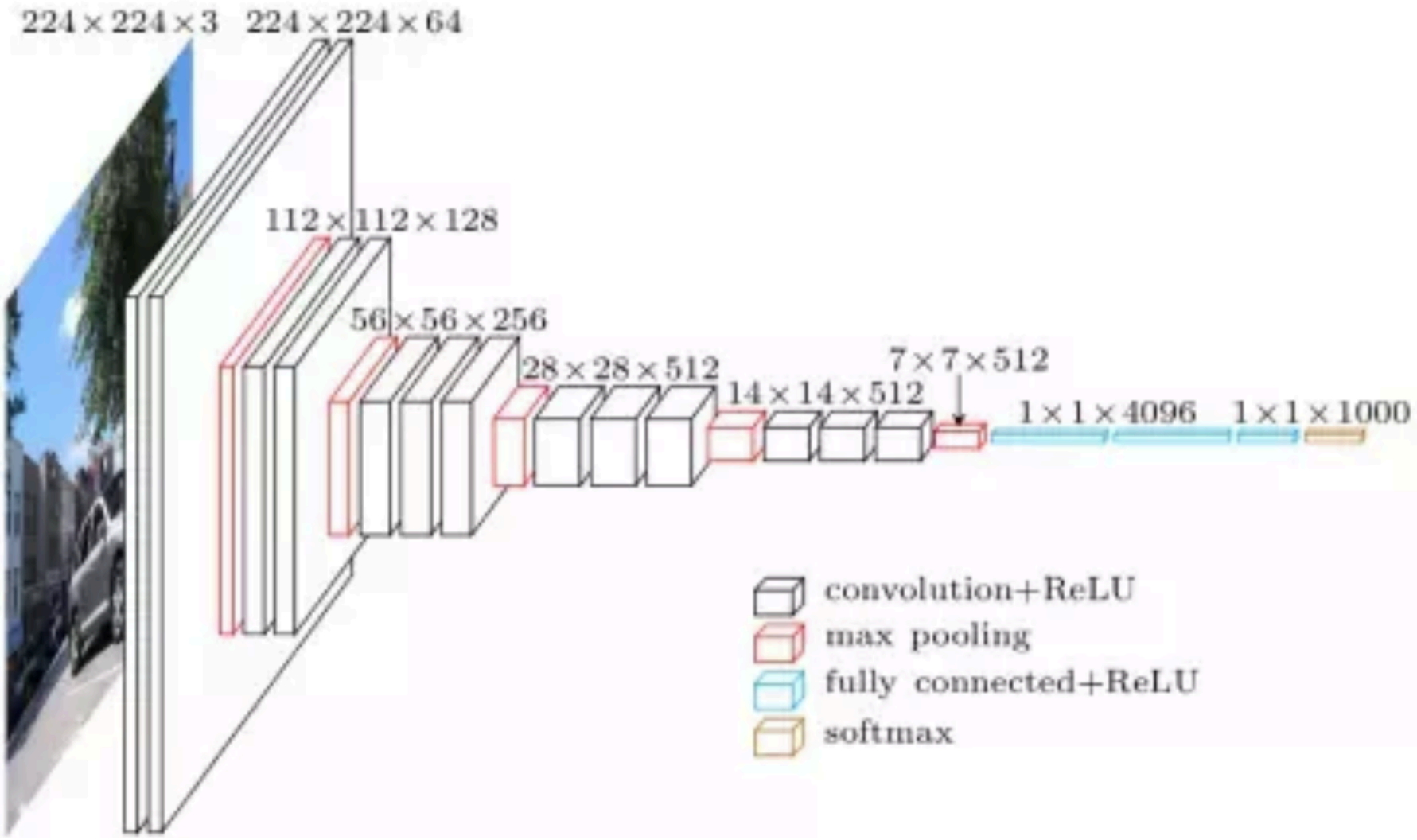
Week 5 Part 2

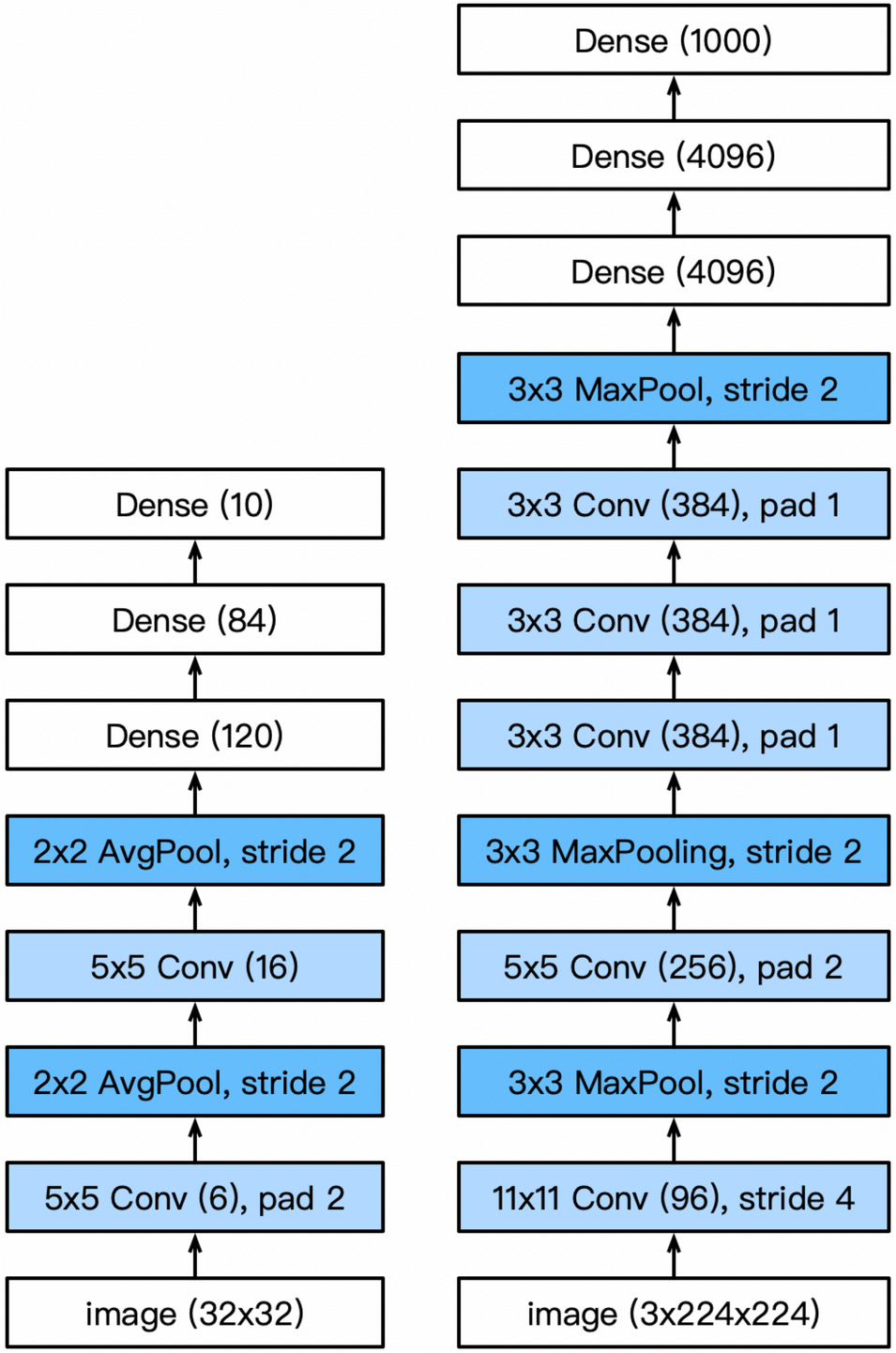
VGG
NiN

VGG


224 × 224 × 3
224 × 224 × 64
112 × 112 × 128
56 × 56 × 256
28 × 28 × 512
14 × 14 × 512
7 × 7 × 512
1 × 1 × 4096
1 × 1 × 1000
convolution+ReLU
max pooling
fully connected+ReLU
softmax

VGG

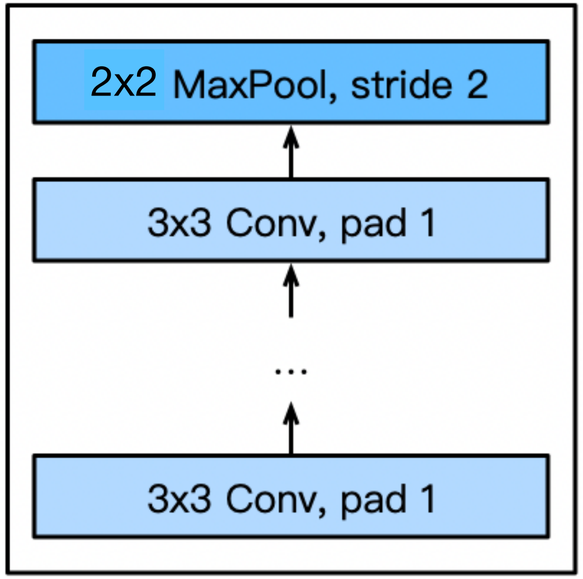
- AlexNet is deeper and bigger than LeNet to get performance
- Go even bigger & deeper?
- Options
  - More dense layers (too expensive)
  - **More** convolutions
  - Group into **blocks**



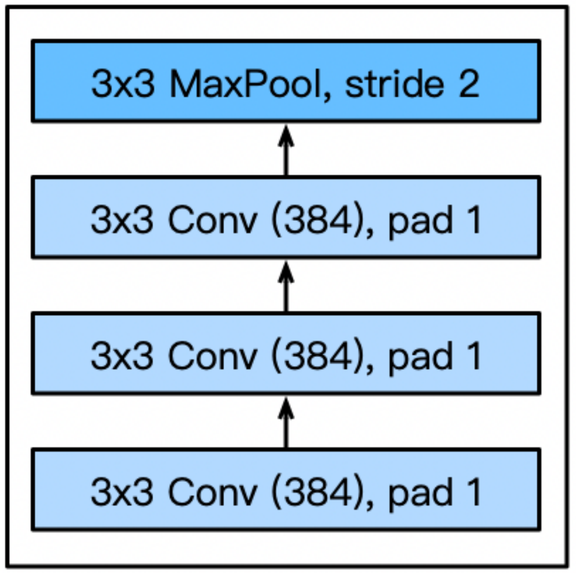
VGG Blocks

- Deeper vs. wider?
  - 5x5 convolutions
  - 3x3 convolutions (more)
  - **Deep & narrow better**
- VGG block
  - *3x3* convolutions (pad 1) **(n layers, m channels)**
  - 2x2 max-pooling (stride 2)

VGG block

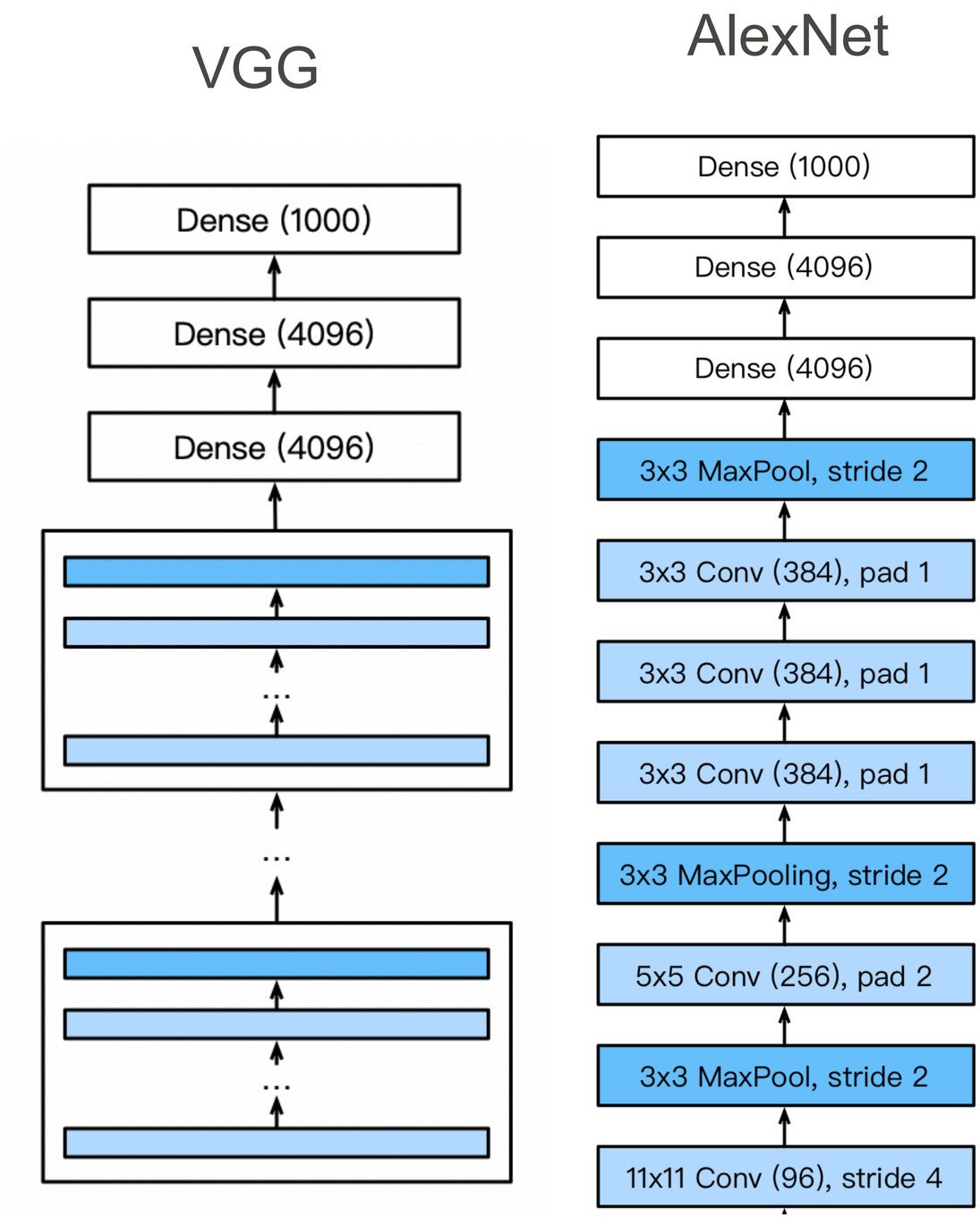


Part of AlexNet



VGG Architecture

- Multiple VGG blocks followed by dense layers
- Vary the repeating number to get different architectures, such as VGG-16, VGG-19, …

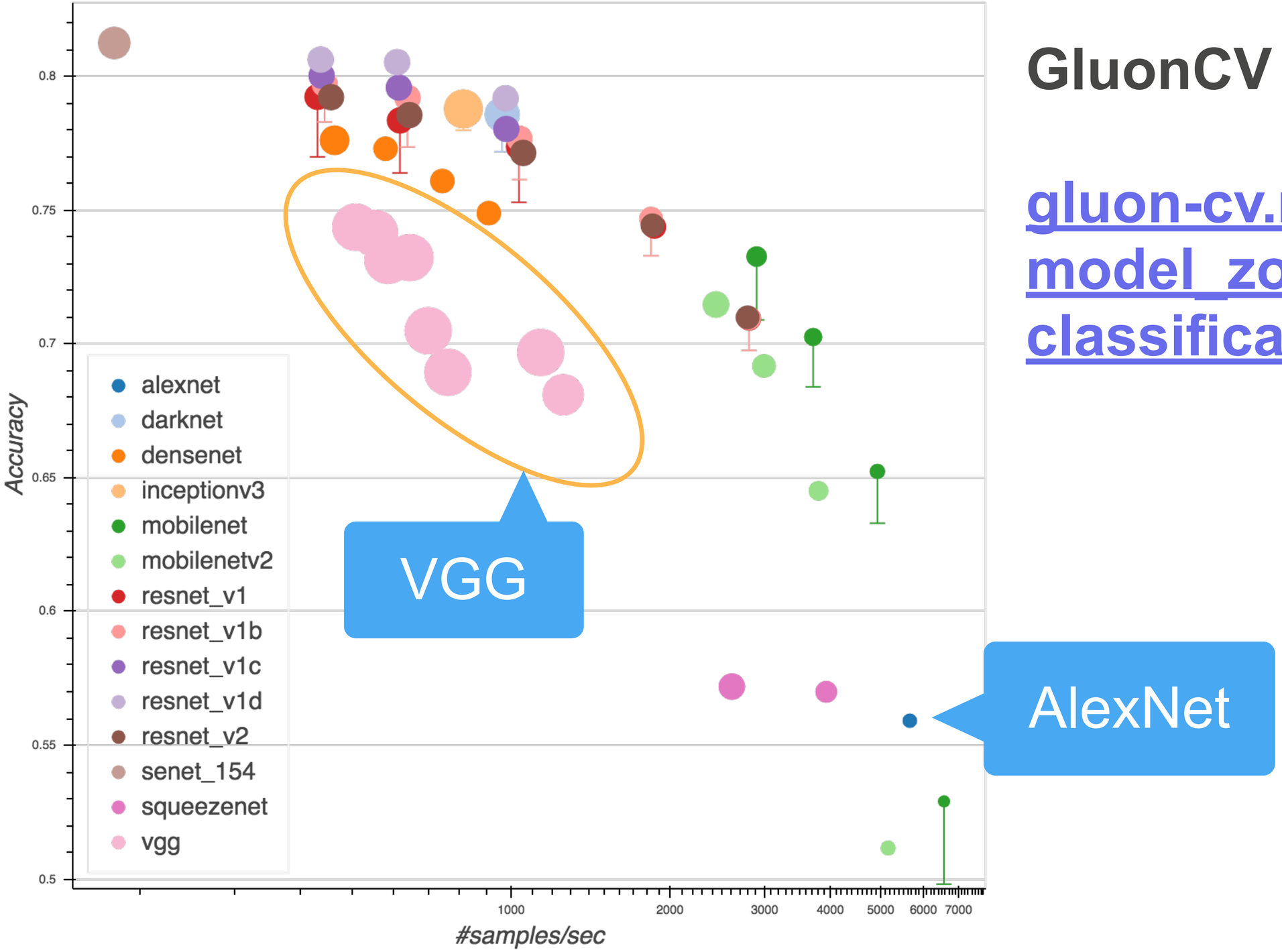


Progress

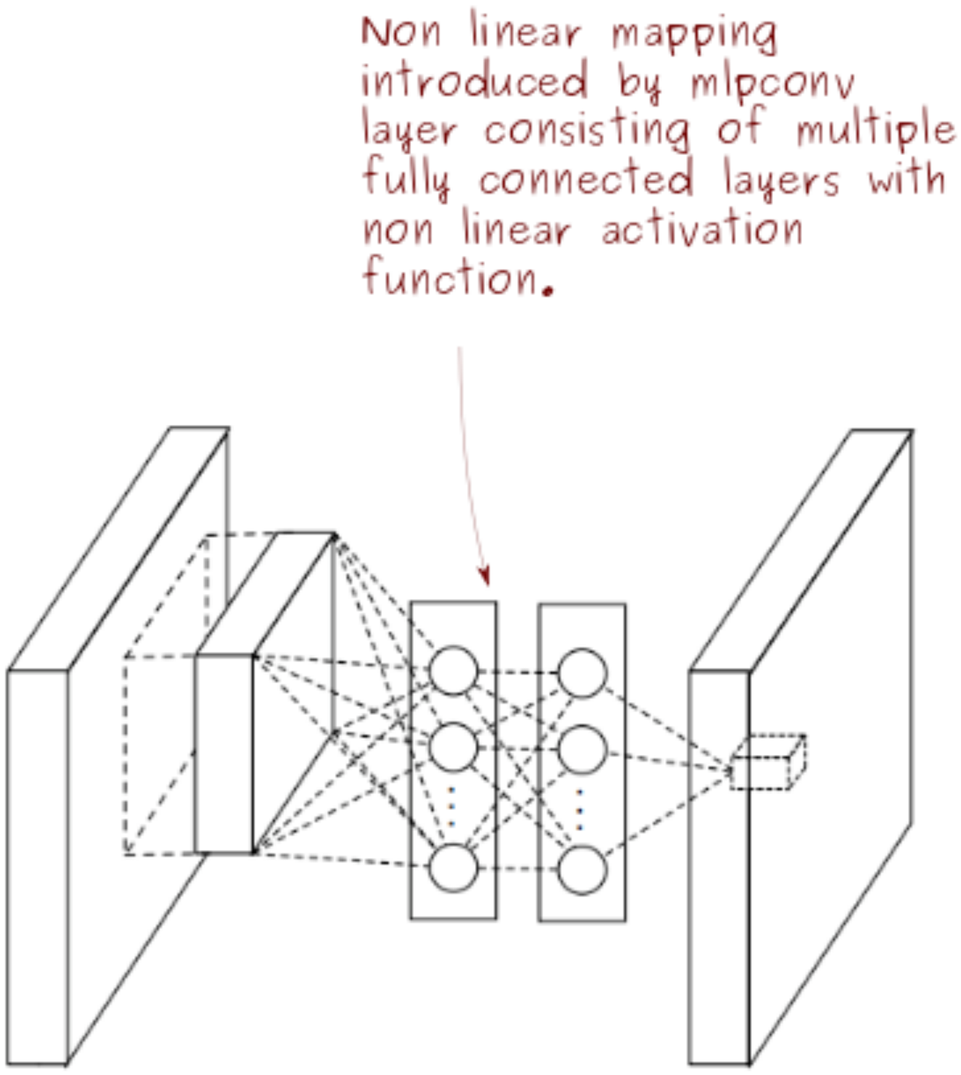
- LeNet-5 (1998)
  - 2 convolution + pooling layers
  - 2 hidden dense layers
- AlexNet (2012)
  - Bigger and deeper LeNet
  - ReLu, Dropout, preprocessing
- VGG (2014)
  - Bigger and deeper AlexNet (repeated VGG blocks)

GluonCV Model Zoo

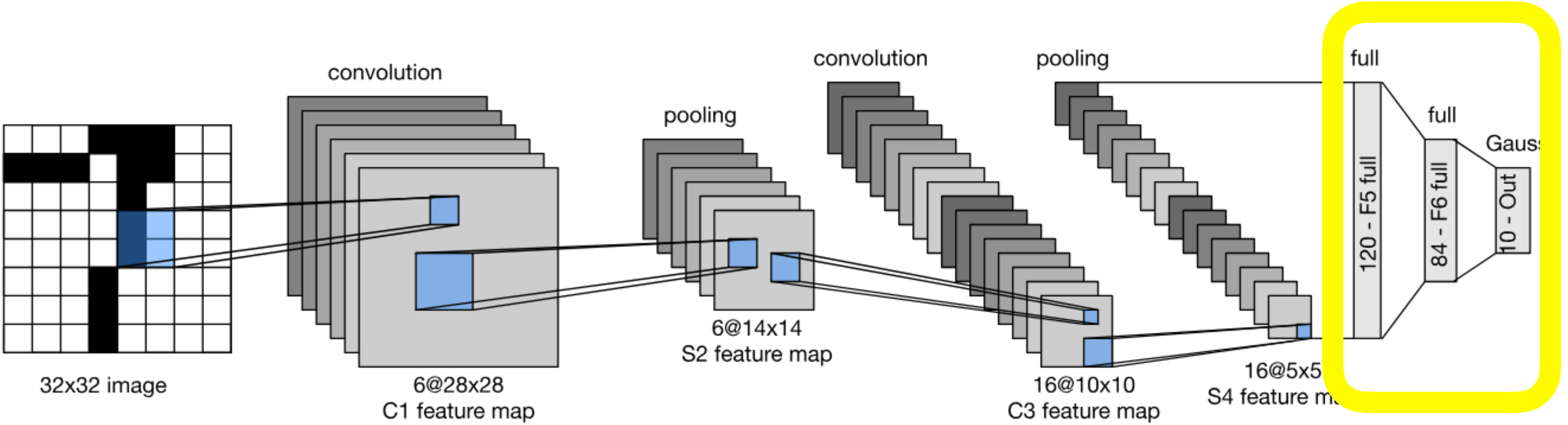
gluon-cv.mxnet.io/model_zoo/classification.html



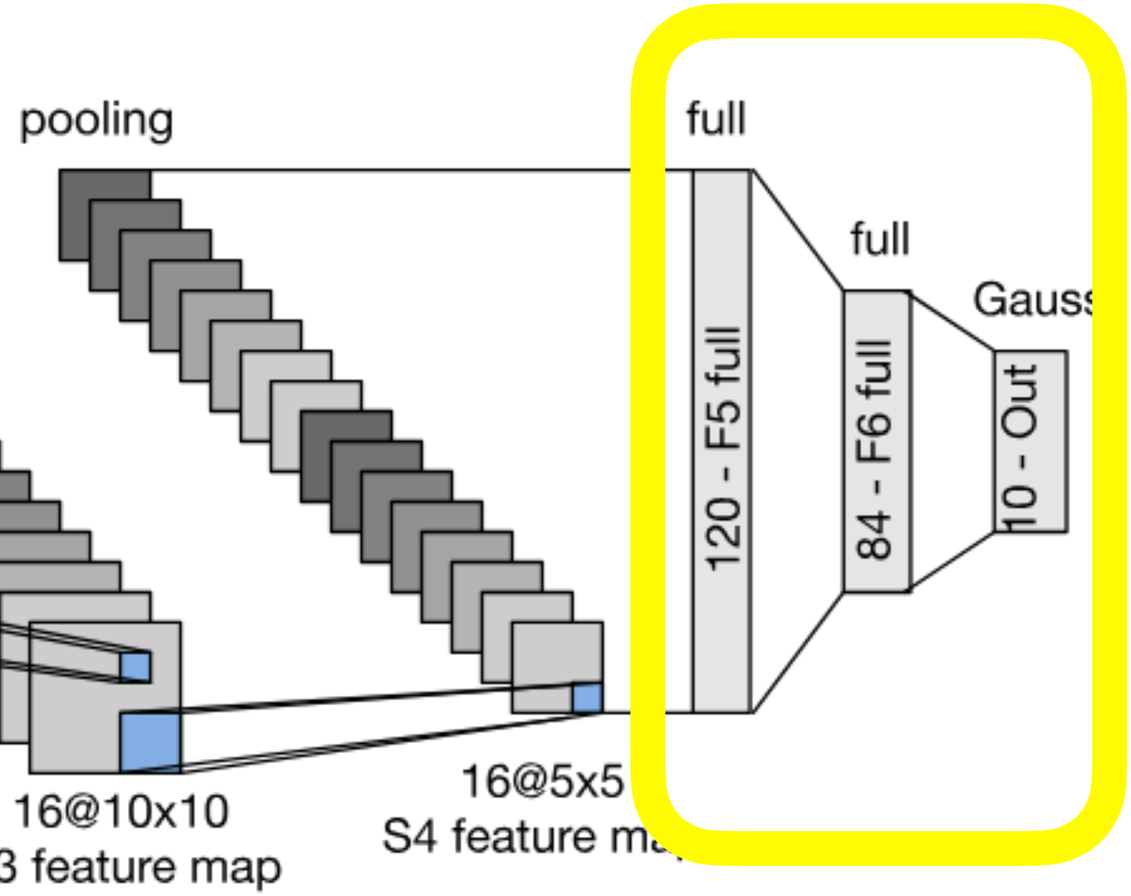
Network in Network



The Curse of the Last Layer(s)



The Last Layer(s)



- Convolution layers need relatively few parameters

$$c_i \times c_o \times k^2$$

- Last layer needs many parameters for n classes

$$c \times m_w \times m_h \times n$$

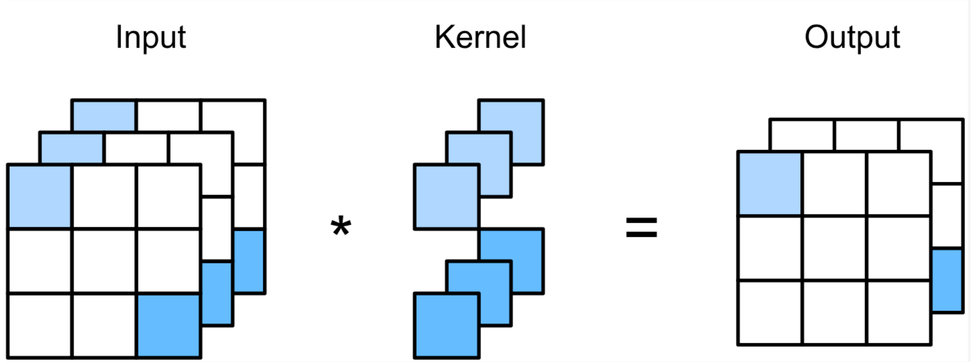
- LeNet 16x5x5x120 = 48k
- AlexNet 256x5x5x4096 = 26M
- VGG 512x7x7x4096 = 102M

VGG parameters

```
sequential1 output shape: (1, 64, 112, 112)
sequential2 output shape: (1, 128, 56, 56)
sequential3 output shape: (1, 256, 28, 28)
sequential4 output shape: (1, 512, 14, 14)
sequential5 output shape: (1, 512, 7, 7)
dense0 output shape:      (1, 4096)
dropout0 output shape:    (1, 4096)
dense1 output shape:      (1, 4096)
dropout1 output shape:    (1, 4096)
dense2 output shape:      (1, 10)
```

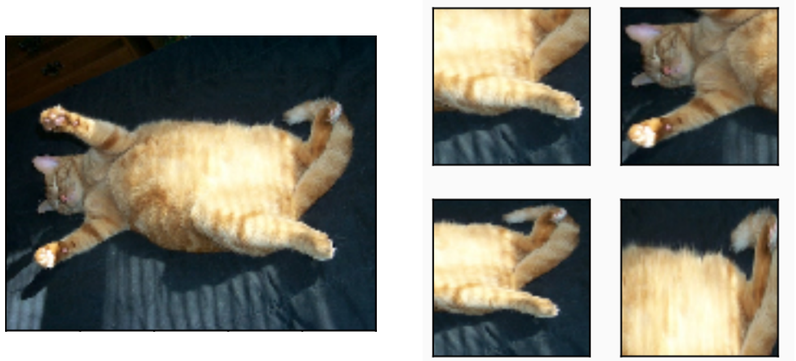
Breaking the Curse of the Last Layer

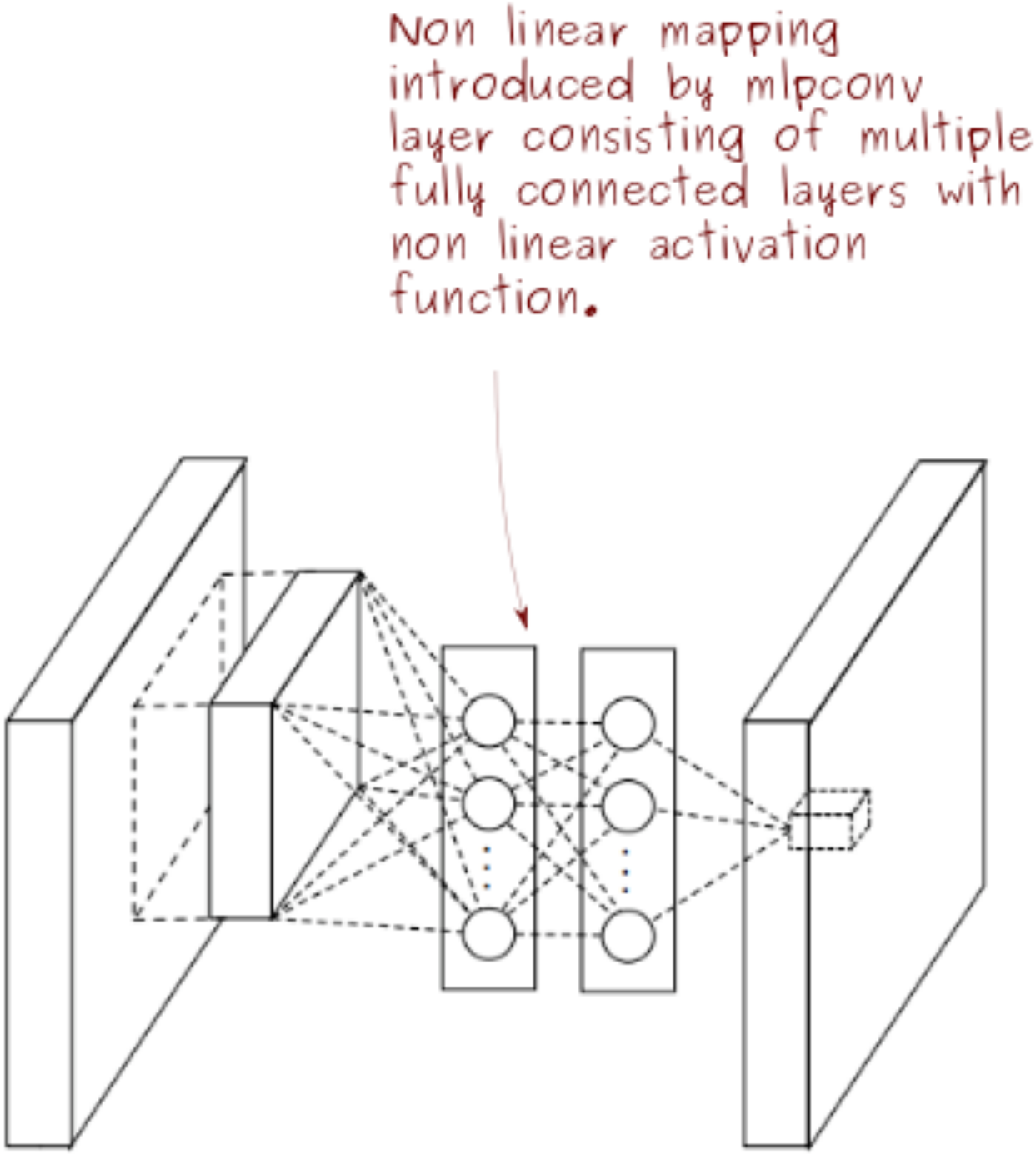
- Key Idea
  - **Get rid of the fully connected last layer(s)**
  - Convolutions and pooling reduce resolution (e.g. stride of 2 reduces resolution 4x)
- Implementation details
  - Reduce resolution progressively
  - Increase number of channels
  - Use **1x1 convolutions** (they only act per pixel)
- **Global average pooling in the end**



What's a 1x1 convolution anyway?

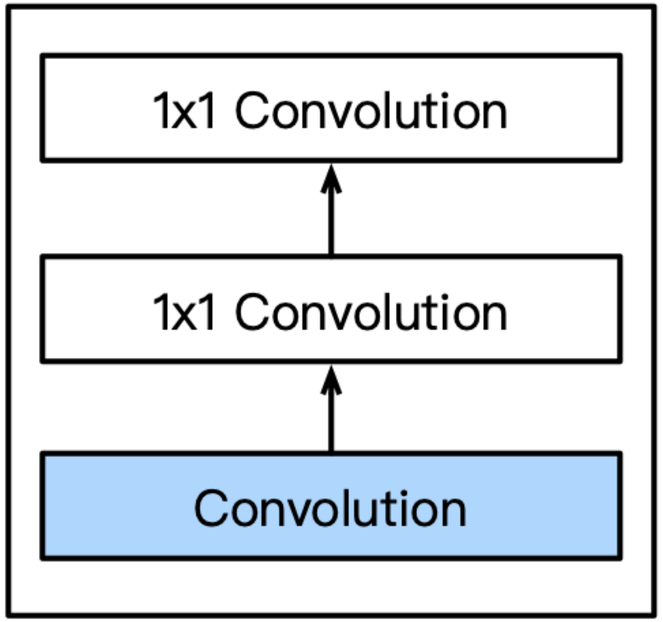
- Extreme case
  1x1 image with n channels
- Equivalent to MLP
- Pooling allows for translation invariance of detection (e.g. 5x5)



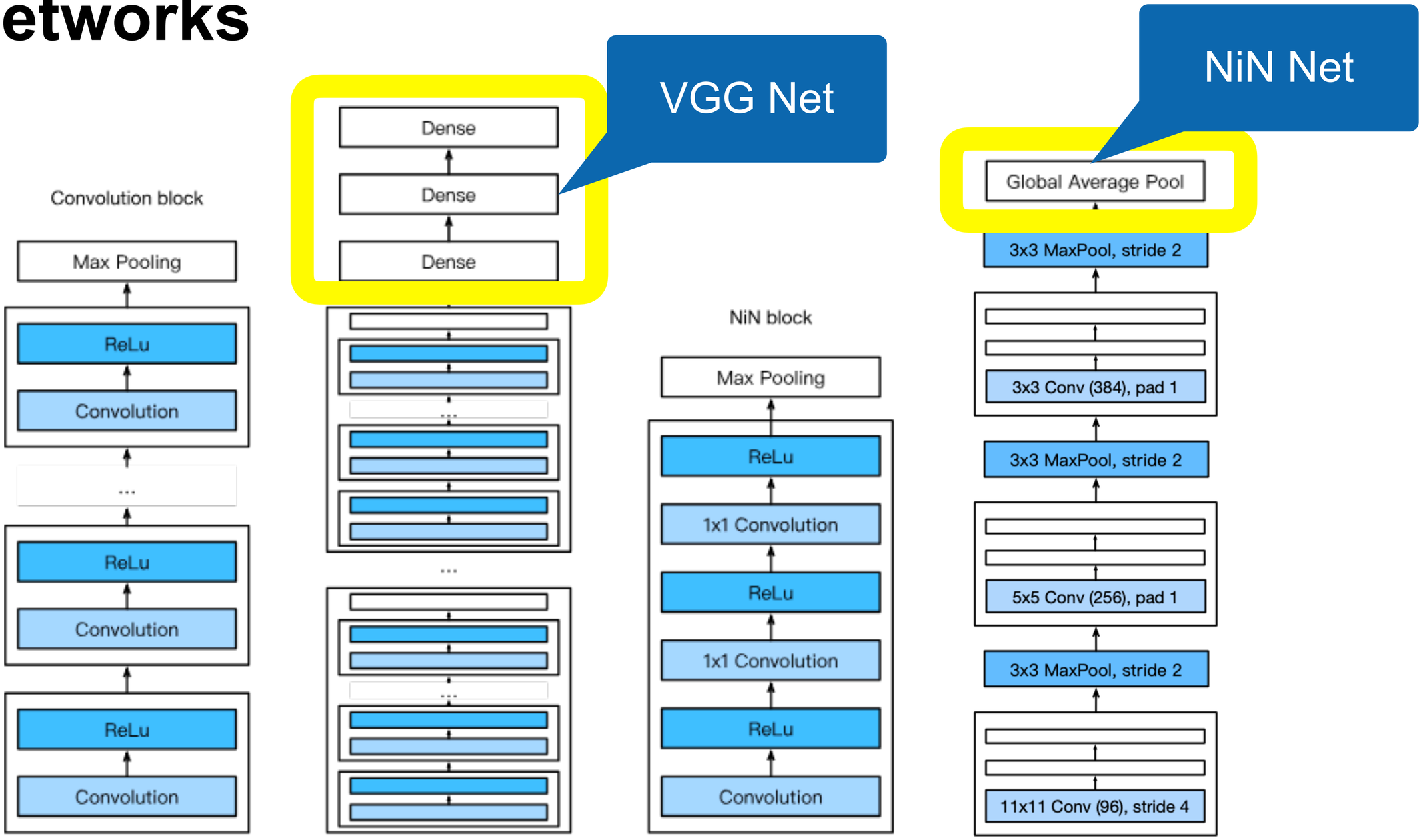


NiN Block

- A convolutional layer
  - kernel size, stride, and padding are hyper-parameters
- Following by two 1x1 convolutions
  - 1 stride and no padding, share the same output channels as first layer
  - Act as dense layers

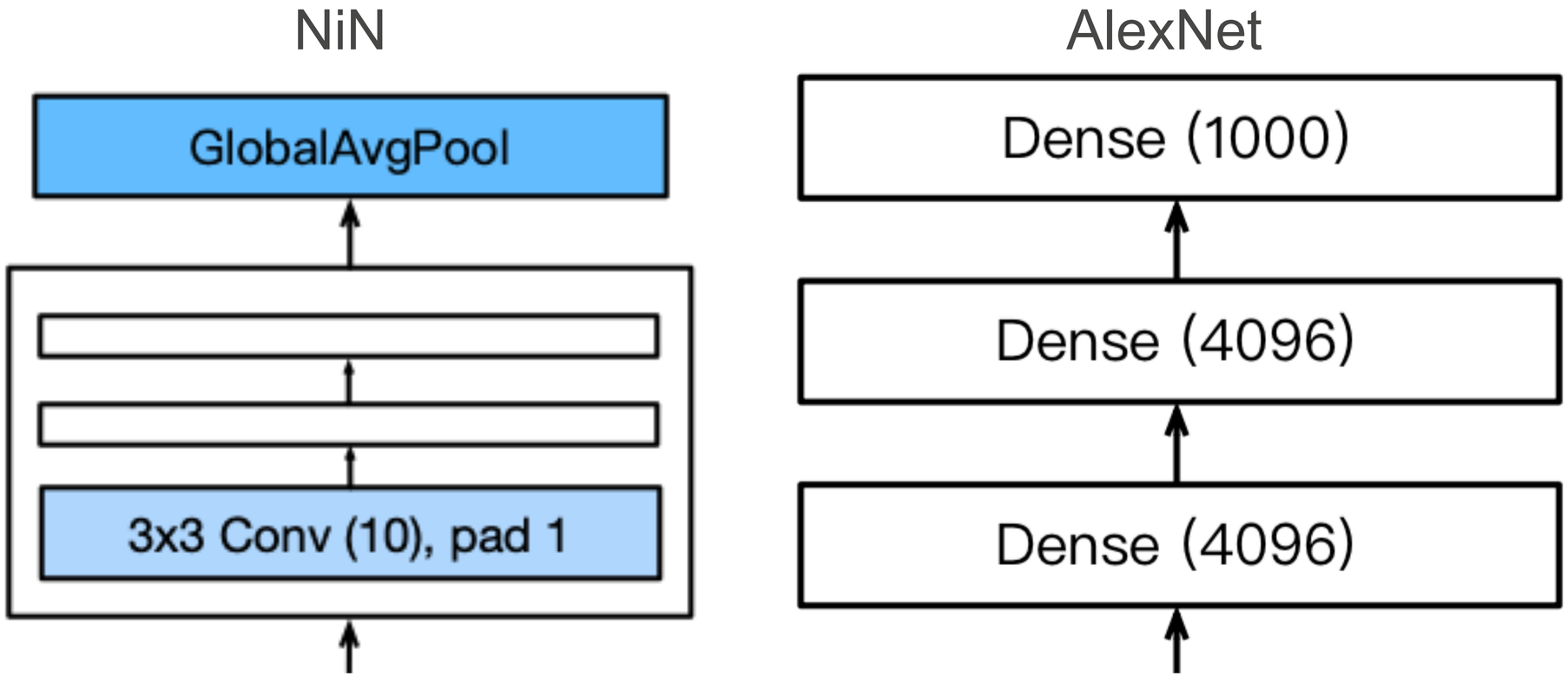


NiN Networks



NiN Last Layers

- Replaced AlexNet's dense layers with a NiN block
- Global average pooling layer to combine outputs



Summary

- Reduce image resolution progressively
- Increase number of channels
- Global average pooling for given number of classes

```
sequential1 output shape: (96, 54, 54)
pool0 output shape:       (96, 26, 26)
sequential2 output shape: (256, 26, 26)
pool1 output shape:       (256, 12, 12)
sequential3 output shape: (384, 12, 12)
pool2 output shape:       (384, 5, 5)
dropout0 output shape:    (384, 5, 5)
sequential4 output shape: (10, 5, 5)
pool3 output shape:       (10, 1, 1)
flatten0 output shape:    (10)
```

NIN dimension reduction

Summary

- **LeNet** (the first convolutional neural network)
- **AlexNet**
  - More of everything
  - ReLu, Dropout, Invariances
- **VGG**
  - Even more of everything (narrower and deeper)
  - Repeated blocks
- **NiN**
  - 1x1 convolutions + global pooling instead of dense